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Knowledge management critical success factors and project management performance outcomes in major construction organisations in Sri Lanka

A case study

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Abstract

Purpose – The purpose of this study is to investigate the influence of knowledge management critical success factors (KM CSFs) on project management performance outcomes in major construction organisations in Sri Lanka. As a result, the significant KM CSFs are prioritised to support KM implementation.

Design/methodology/approach – This study is cross-sectional in nature, adopting the quantitative method of analysis. The research model includes nine KM CSFs (culture, leadership, organisational structure, IT support, T-shaped skills, training, teamwork, performance measurement and benchmarking) and nine items measuring project management performance outcomes. Data were collected from senior, middle and technical staff involved in projects by means of a self-administered survey questionnaire mailed to them.

Findings – The Pearson's correlation coefficient suggests that all of the KM CSFs are adequately correlated except for organisational structure, which has a poor correlation with IT support, T-shaped skills, teamwork and benchmarking. The results of multiple linear regression show that four KM CSFs (T-shaped skills, teamwork, benchmarking and transformational leadership) are significantly and positively related to project management performance outcomes. The remaining five KM CSFs are not significant. The multiple-criteria decision-making analysis reveals that benchmarking, transformational leadership, teamwork and T-shaped skills are the main priority for the construction organisations to consider in implementing KM.

Research limitations/implications – Future studies should consider construction organisations of multiple sizes and include factors beyond organisational control, KM processes and the effects of demographic variables. Longitudinal studies are also imperative to determine the performance impact brought about by KM implementation over a period of time.

Practical/implications – Benchmarking, transformational leadership, teamwork and T-shaped skills should serve as the initial focus to support KM implementation to achieve the desired project management performance outcomes. It is only after these KM CSFs are in place that the construction organisations can concentrate on other factors.

Originality/value – This study contributes to a better understanding of the relationships between the KM CSFs and project management performance outcomes in the construction industry in Sri Lanka, which is still at its infancy stage of KM implementation. Prioritisation of the significant KM CSFs in



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supporting effective KM implementation in the construction organisations further enhances the value of this study.

Keywords Sri Lanka, Knowledge management success factors, Major construction organisations, Project management performance outcomes

Paper type Research paper

1. Introduction

Construction organisations are known to use project management methodologies in planning, executing and monitoring projects (Zhai *et al.*, 2014) to achieve the desired project management performance outcomes (Yeong and Lim, 2015). However, such methodologies have not contributed to the expected outcomes where for the past 70 years (Flyvbjerg *et al.*, 2003), construction projects still suffer from issues relating to cost, time of completion and quality, resulting in dissatisfaction of clients. As a matter of fact, these issues have been found to be the frequent challenges encountered by the construction industry (Handzic and Durmic, 2015), both in the developed and developing countries (Gale and Fellows, 1990).

A different approach is, hence, required to overcome such challenges (Rungasamy and Ghosh, 2002). These challenges had led the organisations to look for a sustainable approach, with knowledge management (KM) implementation identified as a promising one (Enshassi *et al.*, 2016; Nilashi *et al.*, 2015). This is in view of the large number of knowledge workers employed by the construction industry to work in project teams (Kale and Karaman, 2011), along with the increasing utilisation of information technology (IT) to plan, execute and monitor projects, both of which are pre-requisites to KM implementation (Gunasekera and Chong, 2018).

However, successful KM implementation requires a good understanding and proper execution of the various KM processes supported by the KM critical success factors (KM CSFs), and at the same time appreciating the tangible and intangible outcomes brought about by such implementation (Al-Hakim and Hassan, 2012; Nejatian *et al.*, 2013; Zhao *et al.*, 2013). In the construction setting, the KM processes, which enable project team members to create, share and store both tacit and explicit knowledge (Nonaka and Takeuchi, 1995) arising from project planning, execution and monitoring to solve various project-related issues (Muthur *et al.*, 2013) are carried out subconsciously but not documented, leading to knowledge loss. Because of this, knowledge gained from previous projects have not been applied to current projects, leading to project failures (Yeong and Lim, 2010). In addition, the lack of support provided by the KM CSFs (Berraies *et al.*, 2014; Sharimllah Devi *et al.*, 2013) on the implementation of KM has also led to poor performance of projects (Takim and Adnan, 2008).

Idris and Kolawole (2016) found that research on KM CSFs in the construction setting have been given relatively less attention. The available research is fragmented in nature, with no study attempted to investigate all the KM CSFs in the construction industry in a single setting (Gunasekera and Chong, in press). This calls for more empirical research not just to identify but to prioritise on the KM CSFs so that the desired project management performance outcomes can be achieved. Given the fact that the available resources in construction companies and projects are limited (Yong and Mustaffa, 2013), prioritising on the KM CSFs will enable the construction companies to execute the more significant CSFs in project planning, execution and monitoring, leading to the desired outcomes (Al-Zaharani, 2013; Bullen and Rockart, 1981). This are the objectives of this research.

The rest of the article is organised as follows. The next section presents the literature on KM CSFs and project management performance outcomes, which resulted in the

identification of a comprehensive set of variables and a series of hypotheses to be tested in the study. The methodology used is described next, followed by the analysis of data collected. The findings and their implications are discussed before the paper is concluded with future research directions.

2. Literature review

2.1 Knowledge management critical success factors

Rockart (1982) defines critical success factors as the more significant matters in which any organisation or industry should concentrate on based on their limited resources to succeed at any given time. According to Beijerse (2000), KM is the management of information within organisations which is characterised by structure, culture, strategy and systems (Ajmal and Helo, 2010; Bolisani and Scarso, 2015) and organisational leadership (AlShamsi and Ajmal, 2018). These characteristics are generally defined as KM CSFs, which are recognised as the internal issues managed by the construction organisations to mobilise their limited resources to achieve the desired project management performance outcomes (Ajmal *et al.*, 2010; Jennex and Olfman, 2006; Musa *et al.*, 2015). Given the general nature of the characteristics, there is a need to identify a specific set of KM CSFs for the construction industry (Bullen and Rockart, 1981).

For this, Gunasekera and Chong (2018) have carried out a comprehensive review of the literature on KM CSFs in the construction industry using the systematic revision methodology. The review has resulted in the identification of nine KM CSFs. Nevertheless, the work of Gunasekera and Chong (2018) is conceptual in nature where they recommend the KM CSFs to be empirically tested to determine their associations with project management performance outcomes so that appropriate strategies can be formulated to support KM implementation.

Table I shows the KM CSFs and brief explanation of each of the CSFs. The key words are italicised to represent the operationalisation of the KM CSFs. The presence of these KM CSFs is posited to enable construction organisations to achieve the desired project management performance outcomes.

2.2 Project management performance outcomes

The literature suggests that time, cost and quality represent the key measures of project management performance outcomes (Muthur *et al.*, 2013) where KM implementation has been found to exert influence on these measures (Khedhaouria and Jamal, 2015; Laihonen *et al.*, 2015). Time is measured based on project schedules, while cost is measured using project budgets. Quality, on the other hand, is measured using standards and specifications of a project. However, for the sustainability of the construction industry, Lee and Choi (2003) suggest that measures such as market share, revenue growth, profitability and innovation are also important. In addition, Wu and Chen (2014) opine that satisfaction of clients is another important indicator to determine the success of a project.

This study considers nine measures of cost, time, quality, expectation of clients, as well as company success, market share, revenue growth, profitability and innovation as representing the project management performance outcomes.

2.3 Hypotheses

As a result of the literature survey in Table I, as well as the items operationalised to measure project management performance outcomes, below shows the list of hypotheses developed to be tested in this study.

No.	KM CSFs	Explanation
1	Culture	Trust, collaboration, learning and rewards/incentives are recognised as vital in project planning, execution and monitoring (Scarborough *et al.*, 1999). Trust creates *reciprocal faith in the intention and behaviour* of other project members *to work towards organisational interest and goals*, hence enabling a knowledge sharing culture in project management. When the *degree of collaboration between members increases*, members can *support and help each other*, taking *responsibility for failure* to resolve managerial and technical issues in projects (Ajmal *et al.*, 2009; Lee and Choi, 2003). The culture of learning through *formal and informal training* is also necessary to acquire and share project knowledge for decision making (Krough, 1998; Salleh *et al.*, 2013). However, knowledge and information gained are unlikely to be shared if project team members are not *rewarded* for such an action (Samadi, 2015). *Rewards/incentives, as well as trust, collaboration and learning* will lead to better project management performance outcomes (Abualoush *et al.*, 2018)
2		Transformational leadership Transformational leadership is more influential to *motivate and inspire employees* (Crawford, 2005) to achieve common goals of a project through *shared leadership* and *trust* (Azmy, 2012) which influence knowledge sharing (Bass, 1999). Transformational leaders *initiate the acquisition of knowledge* (Politis, 2001) through *active development of leadership skills amongst staff* (Tseng, 2016), *one-to-one mentoring*, *rewards/incentives*, as well as through *informal conversation and meetings* (Choi and Lee, 2002). Project staff with such leaders are likely to work towards common goals, which increases the chance of achieving the desired project management performance outcomes (Ajmal *et al.*, 2017)
3	Organisational structure	The *degree of formalisation and centralisation* influences knowledge sharing and information flow (Berraies *et al.*, 2014). Specifically, the high degree of formalisation and centralisation hinders information flow which subsequently affects performance of projects such as in the case of the construction industry where *decisions from superiors are required* and that *formal or planned basis and written rules and procedures* exist (Lee and Choi, 2003). Nevertheless, a decentralised and less formalised structure is required (Nonaka and Takeuchi, 1995) not only to facilitate KM implementation but also to provide autonomy to project staff to make quick and fast decisions to resolve complex issues (Al-Alawi *et al.*, 2007) relating to cost, time and quality of projects
4	IT support	IT support enables knowledge to be captured, stored and shared between the project team members in a *collaborative manner regardless of time and space* to solve problems and make decisions. *Knowledge can be stored and transferred quickly through communication amongst team members* (Lee and Choi, 2003) using different channels provided by the ICT platform such as e-mails, databases and other means such as internet, workflow, videoconferencing, whiteboard and data mining (Berraies *et al.*, 2014; Storck and Hill, 2000). The stored knowledge allows *easy search, storage and access to information*, which enables *simulation and prediction* (Lee and Choi, 2003) of the possible solutions to critical issues to be carried out, hence leading to the achievement of the desired project management performance outcomes
5	T-shaped skills	People with T-shaped skills are *specialists in their own work and can understand the tasks of others.* They also possess the ability to *communicate with own and other departmental staff and experts who can perform their tasks effectively without regard to environmental changes* (Lee and Choi, 2003). They are regarded as knowledge workers who possess not only functional

(*continued*)

Table I.
KM CSFs and Their explanation in the construction setting

No.	KM CSFs	Explanation
		expertise within their own functions but also other types of knowledge and the ability to integrate them to the different phases and/or activities of a project (Nejatian *et al.*, 2013). Project members with T-shaped skills who can assimilate the different knowledge they possess into new domains of critical knowledge to *make suggestions* and address challenges can contribute to project management performance outcomes (Migdadi, 2005)
6	Training	Continuous training is very important to project staff in light of the nature and demands of projects, which differ from one another, regardless of whether it is conducted *in-house or externally* (Gillani *et al.*, 2018; Zaharias *et al.*, 2001). Although it is very important to share knowledge gained from previous projects, *person-job fit*, skill improvement through training, particularly in *sharing and using KM systems* (Tseng, 2016) will lead to successful KM implementation. In addition, training can overcome the difficulty in recruiting people with multiple skills (Shao *et al.*, 2013) as well as in retaining outstanding project staff in the organisation (Syed-Ikhsan and Rowland, 2004; Tseng, 2016) so that their knowledge and experience can be retained for future projects, which affects project management performance outcomes (Chong *et al.*, 2006)
7	Teamwork	Teamwork is usually more result-oriented (Geraint, 1998) because it sets the environment for project team members to share their knowledge and skills with each other in planning, executing and monitoring project activities to achieve common goals (Conley and Zhang, 2009; Enshassi *et al.*, 2016). To facilitate teamwork, the team needs to be feeling comfortable with *shared leadership*, *spending time with each other*, *joint decision making for better judgment*, as well as *providing support to any team member facing difficulties in performing his/her tasks* (Azmy, 2012) for better project performance
8	Performance measurement	Besides cost, non-financial measures such as time, quality and satisfaction of clients are equally important to represent the project management performance outcomes (Ahamed, 2010). Time-based measures are required to track the milestones set so that necessary actions can be taken to realign project activities which deviated from the original plan. Achieving the required quality is also necessary since poor quality work will lead to increased cost, as well as the additional time required to rectify them. The same goes to the satisfaction of clients. In implementing KM, knowledge activities leading to the project management performance outcomes need to be measured (Carneiro, 2001). Hence, performance measurement should consider the *performance goals and objectives set by management* of the construction organisations which include criteria such as *management, distribution and acceptance of data and information* and the *provision of rewards/recognition* by and for the project teams to *enable performance improvement* (Chong and Choi, 2005)
9	Benchmarking	Best practices can be benchmarked within and outside the construction organisation (Chong *et al.*, 2006) in terms of how *knowledge is generated within and exchanged with external parties*, *acquired from external partners and competitors*, as well as *from organisations outside of the construction industry* (Park, 2006). Benchmarking creates not only a new environment for the application of KM processes but also the application of KM CSFs, hence enabling the construction organisations to achieve the desired project management performance outcomes

Table I.

H1: Organisational culture has a significant influence on the project management performance outcomes.

H2: Transformational leadership has a significant influence on the project management performance outcomes.

H3: Organisational structure has a significant influence on project management performance outcomes.

H4: IT support has a significant influence on project management performance outcomes.

H5: T-shaped skills have a significant influence on project management performance outcomes.

H6: Training has a significant influence on the project management performance outcomes.

H7: Teamwork has a significant influence on project management performance outcomes.

H8: Performance measurement has a significant influence on project management performance outcomes.

H9: Benchmarking has a significant influence on project management performance outcomes.

3. Research methods

3.1 Data collection

Data were collected using a self-administered questionnaire mailed to respondents working for major construction organisations in Sri Lanka. Major construction organisations are defined as those which undertake projects of more than Rs. 600 million in value at any one time.

The population frame of 1,225 comprises 167 top managers, 278 middle managers and 780 technical staff involved in projects. Based on the population frame, a sampling frame of 580 respondents were determined (80 top managers, 190 middle managers and 310 technical staff). The questionnaires were mailed to all of the 580 respondents, with a reminder sent to the Chief Executive Officers of the respected construction organisations three weeks after the initial distribution. A total of 318 responses were received within two-and-a-half-month period from September 11, 2017. One questionnaire was found to be half-filled, while the other was not filled at all, leaving the total of usable responses to 316, representing a response rate of 54 per cent. Based on the population frame of 1,225, Sekaran and Bougie (2009) opine that the sample size required for analysis is 291. However, according to Tabachnick (2007), as a rule of thumb, at least 300 cases are required for factor analysis. Hence, the 316 complete responses are deemed to be adequate.

The top managers, middle managers and technical staff were represented at the rates of 48, 38 and 68 per cent, respectively. The majority of respondents (75 per cent) were males, with most of them belonging to the age category of 25-35 (58 per cent), followed by 36-45 (22 per cent) and 46-55 (15 per cent). Those between the age cohort of 56-65 years only constituted 5 per cent of the total respondents. In terms of education level, those who possess a Bachelor's degree made up the majority (56 per cent), followed by those with technical (21 per cent) and postgraduate qualifications (16 per cent). Most of them have been working for the current organisations for less than 3 years (41 per cent), followed by 3-6 years and 7-10 years, at 32 per cent and 10 per cent, respectively.

3.2 Methods and measures

The survey instrument was developed with nine constructs measuring the KM CSFs and one construct representing project management performance outcomes. To operationalise the constructs and their associated variables (Saunders *et al.*, 2012), various studies were referred to, looking at the fragmented nature of the KM CSFs. Specifically, the variables in the construct measuring culture (trust, collaboration and learning), organisational structure (centralisation and formalisation), as well as items measuring IT support and T-shaped skills were adapted from Lee and Choi (2003). Items measuring rewards were adapted from Samadi (2015), whereas items measuring transformational leadership were taken from various studies (Azmy, 2012; Choi and Lee, 2002; Crawford, 2005; Politis, 2001; Tseng, 2016). Items measuring training were adopted from three studies (Syed-Ikhsan and Rowland, 2004; Tseng, 2016; Zaharias *et al.*, 2001). The constructs of Azmy (2012), Chong and Choi (2005) and Park (2006) were referred to in the development of items measuring teamwork, performance measurement and benchmarking, respectively. The nine items measuring project management performance outcomes were adapted from Lee and Choi (2003) and Muthur *et al.* (2013). All the items were scored on a five-point Likert scale, ranging from 1 = strongly disagree to 5 = strongly agree with a neutral point in between. Appendix 1 shows the list of the survey items.

In addition, five demographic questions were also included in the questionnaire, which consisted of gender, age, educational qualifications, positions and tenure of service in the current organisation. The questions were adopted from Dunk and Jeng (2013).

In addition to multiple linear regression analysis, the multiple criteria decision making (MCDM) analysis was carried out. The MCDM analysis estimates the variation of alternatives according to several methods for deriving at the results of ranking alternatives (Mukhametzyanova and Pamucarb, 2018), hence improving the reliability and reducing the risk of making unsound or subjective decisions. In this study, three MCDM methods were considered: (i) Analytical Hierarchy Process (AHP); (ii) Data Empowerment Analysis (DEA); and (iii) Simple Multi-Attribute Rating Technique (SMART) (Velasquez and Hester, 2013). The AHP method involves the use of pairwise comparison with the dependent variable, whereas the DEA includes the use of multiple inputs and outputs. The SMART method represents the input of all the variables with dependent variable. Relating to the study, the AHP method compares the KM CSFs as pairs with project management performance outcomes, whereas DEA analyses more than one KM CSFs as input with project management performance outcomes as output. The SMART method analyses all the KM CSFs together with project management performance outcomes.

3.3 Validity and reliability

Factor analysis using the principal component analysis and varimax rotation was carried out where all the items were grouped accordingly, suggesting validity in the scales used, with the exception of formalisation. In addition, the reliability of the scales was also met in

view of the Kaiser–Meyer–Olkin (KMO) and Cronbach alpha scores which were greater than 0.70, again with the exception of formalisation. The Bartlett's test of Sphericity also showed a level of significance of all the variables and that the AVE was close or greater than 0.50 for all the variables. Hence, the constructs are valid, except for formalisation which was subsequently dropped from further analysis. Table II shows the details.

4. Findings

Table III shows the mean and standard deviation scores for all the nine KM CSFs. The average mean and standard deviation scores were 3.48 and 0.40, respectively. Trust and

Construct	No of Items	KMO Measure	Bartlett's test of Sphericity *p*-value	Bartlett's test of Sphericity $\chi 2$ (*df*)	AVE	Construct reliability Cronbach's alpha
Culture	22	0.672	0.000	712.324 (231)	0.752	0.905
Trust	6	0.795	0.000	166.618 (15)	0.678	0.903
Collaboration	5	0.755	0.000	92.813 (10)	0.622	0.823
Learning	6	0.873	0.000	149.077 (15)	0.699	0.896
Rewards	5	0.851	0.000	159.028 (10)	0.756	0.917
Transformational leadership	8	0.766	0.000	231.306 (28)	0.455	0.891
Organisational structure	8	0.774	0.000	142.895 (28)	0.629	0.661
Centralisation	5	0.796	0.000	117.201 (10)	0.681	0.879
Formalisation	3	0.607	0.064	7.248 (3)	0.502	0.208
IT support	5	0.869	0.000	147.968 (10)	0.759	0.919
T-shaped skills	5	0.710	0.000	92.688 (10)	0.816	0.796
Training	6	0.875	0.000	189.541 (15)	0.730	0.925
Teamwork	6	0.813	0.000	97.353 (15)	0.564	0.840
Performance measurement	8	0.872	0.000	328.097 (28)	0.724	0.945
Benchmarking	8	0.827	0.000	213.694 (28)	0.614	0.909
Project management performance outcomes	9	0.838	0.000	331.224 (36)	0.796	0.917

Table II.
Results of validity and reliability test

Factor	Mean	SD
Culture		
Trust	3.74	0.58
Collaboration	3.75	0.59
Learning	3.39	0.73
Rewards	3.26	0.81
Transformational Leadership	3.58	0.63
Organisational Structure	3.04	0.61
Centralisation	2.76	0.81
Formalisation	3.26	0.55
IT support	3.61	0.77
T-shaped Skills	3.68	0.60
Training	3.41	0.78
Teamwork	3.93	0.55
Performance measurement	3.39	0.68
Benchmarking	3.70	0.67
Average Mean Factor Scores	*3.48*	*0.40*

Table III.
Mean and standard deviation scores for knowledge management critical success factors

collaboration (two variables of culture) scored above the average mean scores, along with transformational leadership, IT support, T-shaped skills, teamwork and benchmarking. Teamwork scored the highest mean, followed by trust and collaboration, benchmarking, T-shaped skills and IT support. As all of these factors scored above 3.50 out of a five-point Likert scale, the respondents perceived to be important KM CSFs. Similarly, the overall mean score for project management performance outcomes is rather high (mean = 3.82; SD = 0.70). The standard deviation scores were below 1.00, indicating that the respondents have rated consistently on all the elements.

Table IV shows the results of Pearson's correlation coefficient amongst the nine KM CSFs. The highest correlation for each factor, with at least one other factor in the construct, was between 0.300 and 0.577. Although organisational structure was weakly correlated with IT support, T-shaped skills, teamwork, performance measurement and benchmarking, as the KM CSFs do not have a zero or high correlation, the factors fit the model and are thus suitable for further analysis.

The results of multiple linear regression in Table V show that T-shaped skills, teamwork and benchmarking are significant predictors of project management performance outcomes.

Variables	CU	TL	OS	IT	TS	TR	TW	PM	BM
Culture (CU)	1.000								
Transformational leadership (TL)	0.554	1.000							
Organisational structure (OS)	0.318	0.351	1.000						
IT support (IT)	0.300	0.309	0.145	1.000					
T-shaped skills (TS)	0.352	0.282	0.178	0.321	1.000				
Training (TR)	0.545	0.571	0.354	0.414	0.366	1.000			
Teamwork (TW)	0.396	0.486	0.237	0.352	0.303	0.378	1.000		
Performance measurement (PM)	0.464	0.501	0.288	0.305	0.287	0.536	0.358	1.000	
Benchmarking (BM)	0.415	0.557	0.234	0.328	0.329	0.566	0.401	0.577	1.000

Note: Significant at 0.05 level

Table IV. Results of Pearson's correlation coefficient between the knowledge management critical success factors

Model	Unstandardised coefficients B	Unstandardised coefficients Std. Error	Standardised coefficients Beta	T	*p*-value	VIF
(Constant)	0.234	0.291		0.805	0.422	
Culture	0.091	0.074	0.072	1.226	0.221	1.747
Transformational leadership	0.130	0.070	0.119	1.854	0.065	2.058
Organisational structure	0.029	0.053	0.028	0.550	0.583	1.273
IT support	0.013	0.046	0.015	0.283	0.777	1.275
T-shaped skills	0.177	0.059	0.152	3.009	0.003	2.161
Training	0.001	0.059	0.001	0.011	0.991	1.462
Teamwork	0.183	0.069	0.143	2.654	0.008	1.764
Performance measurement	0.026	0.061	0.025	0.422	0.673	1.926
Benchmarking	0.322	0.064	0.310	5.006	0.000	1.318

Note: Dependent variable: project management performance outcomes

Table V. Results of multiple linear regression between knowledge management critical success factors and project management performance outcomes

The variance inflation factor (VIF) values were less than 5, and hence, there is no problem of multicollinearity. Figure 1 shows the residual plot whereby all the residual values were with ± 3 and were at random, whilst Table VI shows that the distribution of the residual was normal. The adjusted R-squared value was 0.369, implying that 37 per cent of the variation in project management performance outcomes can be explained by T-shaped skills, teamwork and benchmarking. In terms of the strength of relationships, the Beta values showed that benchmarking is the strongest predictor, followed by teamwork and T-shaped skills.

Table VII shows the results of stepwise regression analysis. According to Weiers (2010), stepwise regression analysis performs multiple regression several times and eliminate the weakest correlated variables, leaving the best variables that explain the distribution. Besides benchmarking, T-shaped skills and teamwork, transformational leadership is found to be significant as well, with the adjusted R-squared value increased to 0.385, suggesting that approximately 39 per cent of the variation in project management performance outcomes can be explained by the four KM CSFs.

Tables VIII, IX and X show the results of MCDM analysis using the AHP method, DEA method and SMART method, respectively. While benchmarking was positioned in the highest order of priority across all the three methods, the other KM CSFs differed in order of

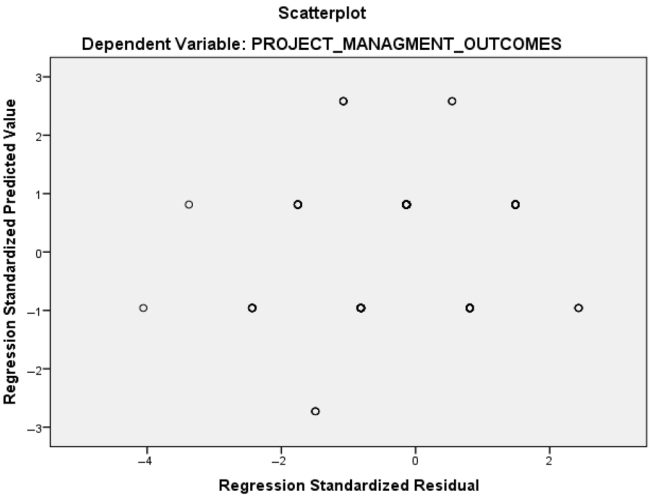


Figure 1.
Standardised residual of project management performance outcomes on the knowledge management critical success factors

Table VI.
Normality test of project management performance outcomes upon the knowledge management critical success factors

	Kolmogorov-Smirnov[a]			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Unstandardised Residual	0.234	316	0.000	0.919	316	0.000

priority, including the significant predictors of project management performance outcomes. Table XI shows that besides benchmarking, two KM CSFs (transformational leadership and teamwork) were ranked rather high and consistent with the results of multiple regression analysis. Although training and performance measurement were ranked high too under the AHP and DEA methods, these KM CSFs were not significant predictors of project management performance outcomes. In contrast, culture, IT support and organisational structure were ranked low across the three methods. On the other hand, although T-shaped skills were ranked rather low under the AHP and DEA methods, its high ranking as

Model	Unstandardised Coefficients B	Unstandardised Coefficients Std. Error	Standardised Coefficients Beta	T	p-value	VIF
(Constant)	0.376	0.270		1.395	0.164	
T Shaped skills	0.199	0.056	0.170	3.532	0.000	1.172
Teamwork	0.202	0.067	0.158	3.007	0.003	1.392
Benchmarking	0.343	0.058	0.330	5.962	0.000	1.552
Transformational leadership	0.179	0.063	0.163	2.852	0.005	1.652

Note: Dependent variable: project management performance outcomes

Table VII. Results of stepwise regression between project management performance outcomes and knowledge management critical success factors

Model	Standardised coefficients Beta	Significance *p*-value	Rank
Benchmarking	0.540	0.000	1
Transformational leadership	0.472	0.000	2
Training	0.424	0.000	3
Teamwork	0.421	0.000	4
Performance measurement	0.405	0.000	5
Culture	0.403	0.000	6
T-shaped skills	0.373	0.000	7
IT support	0.286	0.000	8
Organisational structure	0.247	0.000	9

Table VIII. Results of MCDM analysis using the AHP method

Model	Standardised coefficients Beta	Significance *p*-value	Rank
Benchmarking	0.468	0.000	1
Teamwork	0.405	0.000	2
Transformational leadership	0.394	0.000	3
Performance measurement	0.301	0.000	4
Training	0.296	0.000	5
T-shaped skills	0.294	0.000	6
Culture	0.256	0.000	7
IT support	0.229	0.000	8
Organisational structure	0.216	0.000	9

Table IX. Results of MCDM analysis using the DEA method

illustrated through the use of SMART method and significance to project management performance outcomes deserves attention. It can thus be concluded that benchmarking, transformational leadership, teamwork and T-shaped skills are the priorities of construction organisations in supporting the implementation of KM.

Based on the results, *H2*, *H5*, *H7* and *H9* are supported, whereas *H1*, *H3*, *H4*, *H6* and *H8* are not supported by the data.

5. Discussion and implications

This study has achieved its objective by providing a snapshot of the influences of KM CSFs on project management performance outcomes in the major construction organisations in Sri Lanka. The selection of the major construction organisations in this study was done in deliberation, as they are believed to possess the resources required to implement KM, which can serve to provide a useful reference to other construction organisations.

The overall mean score for project management performance outcomes imply the size and capabilities of the construction organisations in handling larger values of projects and number of clients. Hence, they are in a better position to implement KM through better understanding and prioritisation of the KM CSFs to achieve better project management performance outcomes. Taken together with the overall and individual mean scores of KM CSFs, along with the results of multiple linear regression and MCDM analyses, it is more viable to prioritise on benchmarking, transformational leadership, teamwork and T-shaped

Table X.
Results of MCDM analysis using the SMART method

Model	Standardised coefficients Beta	Significance *p*-value	Rank
Benchmarking	0.310	0.000	1
T-shaped skills	0.152	0.003	2
Teamwork	0.143	0.008	3
Transformational leadership	0.119	0.065	4
Culture	0.072	0.221	5
Organisational structure	0.028	0.583	6
Performance measurement	0.025	0.673	7
IT support	0.015	0.777	8
Training	0.001	0.991	9

Table XI.
Summary of results of MCDM analysis (in terms of ranking of KM CSFs)

Model	Ranking AHP method	DEA method	SMART method
Benchmarking	*1*	*1*	*1*
Transformational leadership	*2*	*3*	*4*
Training	3	5	9
Teamwork	*4*	*2*	*3*
Performance measurement	5	4	7
Culture	6	7	5
T-shaped skills	*7*	*6*	*2*
IT support	8	8	8
Organisational structure	9	9	6

skills, as the construction organisations surveyed are still at the infancy stage of KM implementation.

The significant result of benchmarking is in line with prior studies (Chong *et al.*, 2006; Kale and Karaman, 2011; Luzia *et al.*, 2013; Park, 2006), thus the support for *H9*. As a matter of fact, the mean score of 3.70 also suggests that the respondents perceived benchmarking as one of the most important factors supporting KM implementation. This is not surprising, given the fact that the construction organisations surveyed are still searching for the best method of implementing KM (Kuwaiti *et al.*, 2017). One effective way is through benchmarking the superior practices available to support KM implementation and to achieve the desired project management performance outcomes.

Supporting *H2*, transformational leadership has also been found to have a significant influence on project management performance outcomes, which is consistent with the findings of prior studies (Azmy, 2012; Bass, 1999; Choi and Lee, 2002; Crawford, 2005; Politis, 2001; Tseng, 2016). With the overall mean score of 3.58 and its high ranking on MCDM analysis, the finding implies that top management of construction organisations, as well as leaders of project teams should prioritise by means of enhancing their transformational leadership style to support KM implementation.

Teamwork is another significant predictor of project management performance outcomes, hence the support for *H7*. In fact, this factor scored the highest mean score amongst all the KM CSFs (3.93) and showed a high ranking on the MCDM analysis. This is understandable, as project management involves individuals working in teams in the various stages of projects, that is, technical, procurement, quality control, progress monitoring and construction supervision, to name some, to achieve project goals and objectives. As a matter of fact, teams have been found to outperform in a collaborative environment to achieve common team goals through sharing of knowledge and skills (Ajmal and Koskinen, 2008; Conley and Zhang, 2009). The finding suggests that it is vital to foster shared leadership, camaraderie and trust amongst team members to encourage knowledge to be shared and applied in various phases of projects for decision-making and to resolve issues (Geraint, 1998).

In line with the finding on teamwork, T-shaped skills have been identified as another significant predictor of project management performance outcomes, supporting *H5* and the prior literature (Lee and Choi, 2003; Migdadi, 2005; Nejatian *et al.*, 2013). The finding also corroborates the average mean score (3.68) obtained for this factor. This is not unexpected, given the nature of project teams which comprise knowledge workers (project managers, engineers and technical staff) from different backgrounds who work together to achieve common project goals and deliverables. The finding implies that it is of paramount importance to invest in people who possess such skills and who can contribute to the performance of the team and/or project as a whole.

However, all the dimensions of organisational culture (trust, collaboration, learning and incentives/rewards) and organisational structure (formalisation and centralisation), as well as IT support, training and performance measurement have been found not to be significantly associated with project management performance outcomes. This is despite the higher than average mean scores obtained for trust and collaboration (3.74 and 3.75), as well as IT support (3.61), showing that these KM CSFs are perceived to be important but not seen as contributing to the project management performance outcomes. The remaining KM CSFs, including learning and rewards under the dimension of organisational culture scored below the average mean. The lower than average mean scores, as well as the insignificant associations between culture, organisational structure, training, performance measurement and project management performance outcomes fetch two possibilities. One is on the extent

of understanding of the construction organisations on these KM CSFs. The second possibility, which is more peculiar, is that these KM CSFs should not be prioritised at this point of time. The results from the benchmarking exercise could enable the organisations to see a wider spectrum of what constitutes KM CSFs.

As the stepwise regression analysis have resulted in the retention of the four KM CSFs which was confirmed by the MCDM analysis, benchmarking, transformational leadership, teamwork and T-shaped skills should be given priority by the construction organisations in mobilising their resources to support KM implementation. The finding on benchmarking suggests that the construction organisations should conduct benchmarking exercise to improve their understanding on the scope and execution and identifying superior practices in terms of successful KM implementation and the support required from various KM CSFs. The results of Pearson's correlation coefficient suggest several KM CSFs that can be benchmarked upon, such as the promotion of teamwork and adoption of transformational leadership, possibly including performance measurement, training and a knowledge-friendly culture. To begin with, the construction organisations could identify areas of benchmark and start from within the organisations before looking outside (Chong *et al.*, 2006). By looking at superior practices, they do not have to reinvent the wheel and cost is saved. External benchmarking is carried out in areas which the organisations are lacking.

In addition, top management should empower and delegate responsibilities to project managers to maintain the motivational level of project team members to set collective goals and to engage them in KM activities. In other words, top management must exercise effective transformational leadership style, specifically the consulting and delegating type of leadership, to motivate senior and junior project staff to acquire and share relevant knowledge by institutionalising appropriate rewards and incentives (Ajmal *et al.*, 2010; Choi and Lee, 2002). Management must constantly ensure that there is synergy within and between teams to achieve the common goals set rather than concentrating on individualised and competitive goals (Enshassi *et al.*, 2016). Rewarding the project team as a whole rather than on individual members for using knowledge processes to achieve the desired project management performance outcomes should be emphasised. In addition, the results of Pearson's correlation coefficient also suggest several KM CSFs pertaining to top management support that should be considered in the future, such as training and performance measurement once the construction organisations have started to implement KM.

At the same time, it is also imperative to promote trust and collaboration where encouragement is provided to every project team member to provide input and suggestions for decisions to be taken and for remedial actions to be proposed to resolve any issue facing projects (Azmy, 2012). Continuous exchange of knowledge, skills and expertise between team members, particularly from those with richer expertise in technical and project management skills to other team members is required for learning to take place and for knowledge to be acquired and created. In this respect, top management of the construction organisations plays an important role in promoting teamwork through the adoption of transformational leadership.

As training is not the priority at this point of time, investment in people with T-shaped skills is important, as they have direct effects on the KM processes and the ability to identify and pre-empt possible situations and/or issues facing projects (Gunasekera and Chong, in press). It is recommended that the construction organisations identify candidates with the competencies required, devise an attractive recruitment plan and allocate them to the right composition of teams. For existing staff, a proper system needs to be put in place to

encourage them to understand, communicate and provide suggestions to the tasks of other team members, teams or departments (Lee and Choi, 2003).

6. Conclusion

This paper has achieved its objectives by identifying the influence of KM CSFs on project management performance outcomes in major construction organisations in Sri Lanka. From the research perspective, it has contributed to knowledge in this area, whereas from the practical perspective, some recommendations have been made in light of the findings. It is hoped that the research sheds light to the construction organisations on the KM CSFs to be prioritised to support KM implementation. Indeed, many studies have insisted that KM can be implemented when the necessary environment is in place (Borowska, 2017; Farzin *et al.*, 2014; Radujkrovic and Sjekavica, 2017; Savoiu and Tudoroiu, 2017; Yu *et al.*, 2017).

This study is not without limitations. Data were collected from senior, middle and technical staff of major construction organisations only without considering the other types of organisations. Future studies should consider the different types of organisations. In addition, the adjusted R-squared values imply that future studies should consider other factors, including those beyond organisational control such as weather conditions, social, political, economic, industry and construction technology influences (Gunasekera and Chong, 2018; Yeong and Lim, 2010), as well as integrating them with the KM processes. The possible mediating effects of organisational culture (Ajmal and Kakale, 2009) should be investigated too. In addition, the effect of demographic factors such as gender, age, education level, position and tenure of service should be considered in future studies to further inform on the appropriate composition of team members. The inclusion of these factors will enable the construction organisations to formulate appropriate strategies to successfully implement KM to achieve the desired project management performance outcomes. As KM implementation requires time, a longitudinal study is also proposed to capture the finer details.

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Appendix 1

Variables	Items
Culture	
Trust	Our company members. . .
TR1	are generally trustworthy
TR2	have reciprocal faith in the intention and behaviours of other members
TR3	have reciprocal faith in the behaviours of others to work towards organisational goal
TR4	have reciprocal faith in the behaviours of others to work towards organisational goal
TR5	have reciprocal faith in the decision of others towards organisational interest than individual interest
TR6	have relationship based on reciprocal faith
Collaboration	
CO1	Our organisation members are satisfied with the degree of collaboration
CO2	Our organisation members are supportive of each other
CO3	Our organisation members are helpful
CO4	There is a willingness to collaborate across organisational units within our organisation
CO5	There is a willingness to accept responsibility for failure
Learning	Our company. . .
LE1	provides various formal training programmes related to the performance of our duties
LE2	provides opportunities for informal individual development other than formal training such as work assignment and job rotation
LE3	encourages people to attend seminars, symposia and so on
LE4	provides various programmes such as clubs and community gathering
LE5	members are satisfied by the contents of job training
LE6	members are satisfied with the self-development programmes
Rewards/Incentives	
RE1	It is more likely that I will be given a pay rise or promotion if I finish a large amount of work
RE2	It is more likely that I will be given a pay rise or promotion if I do a high-quality work
RE3	Getting work done quickly increase my chances of a pay rise or promotion
RE4	Getting work done on time is rewarded with high pay
RE5	When I finish my job on time, my job is more secured
Transformational Leadership	
TL1	I feel comfortable with the concept of shared leadership
TL2	Our organisational leaders motivate employees to share knowledge
TL3	Our organisational leaders build up trust amongst employees to share knowledge
TL4	Our organisational leaders promote initiatives to acquire knowledge
TL5	Our organisation actively develops leadership skills of our staff
TL6	In our organisation, knowledge is acquired by one-to-one mentoring
TL7	In our organisation, informal conversations and meeting are used for knowledge sharing
TL8	Our organisation provides rewards and incentives for sharing knowledge

(*continued*)

Table AI.
List of survey items

Variables	Items
Organisational structure	
Centralisation	Our company members. . .
CE1	can take actions without a superior
CE2	are encouraged to make their own decisions
CE3	do not need to refer to someone else
CE4	do not need to ask their supervisors before taking actions
CE5	can make decisions without approval
Formalisation	In our company. . .
FO1	there are many activities that are not covered by some formal procedures
FO2	contacts with organisational members are made on a formal or planned basis
FO3	rules and procedures are typically written
IT Support	Our company. . .
IT1	provides IT support for collaborative works regardless of time and place
IT2	provides IT support for communication amongst organisational members
IT3	provides IT support for searching for and accessing necessary information
IT4	provides IT support for simulation and prediction
IT5	provides IT support for systematic storing of data and information
T-Shaped Skills	Our company members. . .
TS1	can understand not only their own tasks but also the tasks of others
TS2	can make suggestions about the tasks of others
TS3	can communicate well not only with their departmental members but also with members of other departments
TS4	are specialists in their own parts
TS5	can perform their own tasks effectively without regard to environmental changes
Training	Our organisation. . .
TE1	places people at the right job position
TE2	provides training for sharing of knowledge
TE3	provides continuous training programme within the organisation
TE4	provides continuous training programme outside the organisation
TE5	facilitates us to use knowledge management systems
TE6	is able to retain outstanding staff
Teamwork	
TW1	I feel comfortable with the concept of shared leadership
TW2	I feel comfortable with the decision-making process within the team
TW3	I spend time with team members to clarify the expectations of the team
TW4	Team exercises good judgement during decision-making process
TW5	Team members provide input/thoughts throughout the project
TW6	I help my team whenever anyone has difficulties in performing tasks
Performance measurement	Our company employs a procedure to measure. . .
PM1	distribution of knowledge within the organisation
PM2	amount of reports generated on knowledge activity by employees
PM3	number of relationships established due to knowledge systems and networking

(*continued*)

Table AI.

Variables	Items
PM4	number of employees accepting knowledge activity as part of their daily work
PM5	changes of job performance due to proper management of knowledge in place
PM6	performance of target activities to previously set baseline
PM7	job performance data and information
PM8	actual performance improvement and reward/recognition
Benchmarking	Our company has processes for. . .
BM1	generating new knowledge from existing knowledge
BM2	using feedback from past experience to improve future projects
BM3	exchanging knowledge with external partners
BM4	acquiring knowledge about new products and services within our industry
BM5	acquiring knowledge about competitors within our industry
BM6	benchmarking performance amongst employees and departments
BM7	identifying and upgrading best practices
BM8	encouraging employees to benchmark best practices of other organisations
Project management performance outcomes	Our company's project management processes generally allow us to. . .
PO1	meet project cost
PO2	meet project schedules
PO3	meet deliverable quality
PO4	meet customer expectations
PO5	Compared with competitors, our company. . . is more successful
PO6	has a greater market share
PO7	is growing fast
PO8	is more profitable
PO9	is more innovative

Table AI.

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